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MH1200 Linear Algebra I

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1. Linear Independence

Definition Linear Dependence

Definition 1

A list of vectors $\overrightarrow{w_i} \in S$ is linearly dependent iff some $\vec{v} \in S$ is a linear combination of the vectors $\overrightarrow{w_i} \in S$.

Definition Linear Dependence, Alternative Definition

Definition 2

Fix some Euclidean space $\mathbb{R}^n$. Consider some finite list of vectors in tis $\mathbb{R}^n$:

$$\overrightarrow{v_1}, \cdots, \overrightarrow{v_m} \in \mathbb{R}^n$$

This list of vectors is said to be linearly dependent if the zero vector can be expressed as a linear combination of the vectors on the list, with at least one of the coefficients being non-zero.

Example Standard Basis Vectors

Example 3

Show that the vectors $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$ are linearly independent.

Solution:

Consider the following equation:

$$c_1(1,0,0) + c_2(0,1,0) + c_3(0,0,1) = \vec{0}$$

We want to show tha the only solution to this equation is the trivial solution.

Notice that the equation above simplifies to:

$$(c_1, c_2, c_3) = \vec{0}$$

So we immediately have $c_1 = c_2 = c_3 = 0$.

Thus, any solution must satisfy the equation above, but this implies that the only solution to the original equation is the trivial solution.

Therefore, the set of vectors is linearly independent.